

PAPER_290: Crab SNR DPM Vacuum Dilution — $a_{\text{DPM}}(t) \propto r(t)^{-3}$ in Expanding Pulsar Wind Nebula

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Series: UQFF Whitepaper Series — Session 82

Module: CRAB_RESONANCE_UQFF_MODULE.cpp (24th C++ module — FIRST UQFF Pulsar Wind Nebula module)

Date: March 2026

Abstract

This paper establishes the first UQFF module in which the Dirac-Plasmodic Momentum (DPM) vacuum seed acceleration is explicitly time-dependent, evolving as the inverse cube of the expanding Supernova Remnant (SNR) radius. The Crab Nebula (M1, SN 1054 CE, ~ 2 kpc) is the reference system: a pulsar-wind nebula whose shock-driven filaments expand at $v_{\text{exp}} = 1.5 \times 10^6$ m/s. We derive the DPM dilution law, compute the current gravity signal, and show that the DPM vacuum coupling diminishes by a factor $D = 6.69$ over the 971-year life of the nebula.

1. System Parameters

Parameter	Symbol	Value	Notes
Remnant mass	M	9.149×10^{30} kg	4.6 M_{sun}
Initial radius	r_0	5.2×10^{16} m	~ 1.7 pc current
Expansion velocity	v_{exp}	1.5×10^6 m/s	Crab shock front
DPM frequency	f_{DPM}	1×10^{12} Hz	THz resonance
Plasmodic vacuum energy	E_{vac}	7.09×10^{-36} J/m ³	UQFF universal
Current proxy	I_{curr}	1×10^{21} A	Pulsar wind current
Vortical area	A_{vort}	3.142×10^8 m ²	PWN proxy
$\omega_1 - \omega_2$	$\Delta\omega$	2×10^{-3} rad/s	Differential spin
Age at current epoch	t_{age}	3.064×10^{10} s	971 years

2. Core Physics: Dynamic Volume Dilution

2.1 DPM Force (time-independent)

$$F_{\text{DPM}} = I_{\text{curr}} \times A_{\text{vort}} \times (\omega_1 - \omega_2) = 1 \times 10^{21} \times 3.142 \times 10^8 \times 2 \times 10^{-3} = 6.284 \times 10^{26} \text{ N}$$

2.2 Dynamic System Volume

The SNR expands as a sphere of radius $r(t) = r_0 + v_{\text{exp}} \cdot t$:

$$V_{\text{sys}}(t) = \frac{4}{3}\pi (r_0 + v_{\text{exp}} \cdot t)^3$$

This is the **FIRST UQFF module** where V_{sys} is a function of time rather than a fixed system volume.

2.3 DPM Seed Acceleration (time-dependent)

$$a_{\text{DPM}}(t) = \frac{F_{\text{DPM}} \cdot f_{\text{DPM}} \cdot E_{\text{vac}}}{c \cdot V_{\text{sys}}(t)} \propto \frac{1}{r(t)^3}$$

2.4 Numerical Evaluation

At t = 0 (SN 1054 CE explosion):

$$V_0 = \frac{4}{3}\pi(5.2 \times 10^{16})^3 = 5.889 \times 10^{50} \text{ m}^3$$

$$a_{\text{DPM}}(t=0) = \frac{6.284 \times 10^{26} \times 10^{12} \times 7.09 \times 10^{-36}}{3 \times 10^8 \times 5.889 \times 10^{50}} = 2.521 \times 10^{-56} \text{ m/s}^2$$

At t = 971 yr = 3.064×10¹⁰ s (current epoch):

$$r(971 \text{ yr}) = 5.2 \times 10^{16} + 1.5 \times 10^6 \times 3.064 \times 10^{10} = 9.796 \times 10^{16} \text{ m}$$

$$V(971 \text{ yr}) = \frac{4}{3}\pi(9.796 \times 10^{16})^3 = 3.936 \times 10^{51} \text{ m}^3$$

$$a_{\text{DPM}}(971 \text{ yr}) = \frac{6.284 \times 10^{26} \times 10^{12} \times 7.09 \times 10^{-36}}{3 \times 10^8 \times 3.936 \times 10^{51}} = 3.772 \times 10^{-57} \text{ m/s}^2$$

3. DPM Dilution Law

$$D = \frac{a_{\text{DPM}}(t=0)}{a_{\text{DPM}}(971 \text{ yr})} = \frac{V(971 \text{ yr})}{V_0} = \left(\frac{r(971 \text{ yr})}{r_0} \right)^3 = \left(\frac{9.796}{5.2} \right)^3 = (1.884)^3 = 6.69$$

Epoch	r(t) [m]	V_sys(t) [m ³]	a_DPM(t) [m/s ²]
t = 0 (SN 1054 CE)	5.200×10 ¹⁶	5.889×10 ⁵⁰	2.521×10 ⁻⁵⁶
t = 100 yr	6.674×10 ¹⁶	1.242×10 ⁵¹	1.194×10 ⁻⁵⁶
t = 500 yr	8.566×10 ¹⁶	2.629×10 ⁵¹	5.641×10 ⁻⁵⁷
t = 971 yr (now)	9.796×10 ¹⁶	3.936×10 ⁵¹	3.772×10 ⁻⁵⁷
t = 2000 yr	1.154×10 ¹⁷	6.432×10 ⁵¹	2.308×10 ⁻⁵⁷
t = 10000 yr	1.520×10 ¹⁷	1.470×10 ⁵²	1.009×10 ⁻⁵⁷

Dilution factor over 971 years: D = 6.69

4. THz Cascade Amplification (Crab-Specific)

The Crab-specific expansion velocity yields a dramatically enhanced THz amplification factor:

$$\Gamma_{\text{THz}} = \frac{10 \cdot f_{\text{THz}} \cdot v_{\text{exp}}}{c} = \frac{10 \times 10^{12} \times 1.5 \times 10^6}{3 \times 10^8} = 5.0 \times 10^{10}$$

Comparison with RSC module (Session 81): - RSC $v_{\text{exp}} = 1 \times 10^3 \text{ m/s} \rightarrow \Gamma = 3.33 \times 10^7$ - Crab $v_{\text{exp}} = 1.5 \times 10^6 \text{ m/s} \rightarrow \Gamma = 5.0 \times 10^{10}$ (**1500× larger**)

This makes the Crab the highest Γ_{THz} value among all current UQFF modules. The SNR shock expansion directly amplifies the THz cascade chain by over three orders of magnitude relative to neutron star scale systems.

5. Full $g_{\text{Crab}}(t, B)$ Equation

The complete UQFF gravity formula is:

$$g_{\text{Crab}}(t, B) = \left[\sum_i a_i(t) \right] \times \left(1 - \frac{B}{B_{\text{crit}}} \right) \times (1 + f_{\text{TRZ}})$$

where the sum includes: $a_{\text{DPM}}(t)$, a_{THz} , a_{aether} , $a_{\text{u_g4i}}$, a_{quantum} , a_{fluid} , a_{osc} , a_{exp} .

At $t = 971$ yr, $B = 1 \times 10^{-8}$ T: - $\text{SCm} = 1 - 10^{-8}/10^{11} \approx 1.000$ (no quench) - $g_{\text{Crab}} \approx$ dominated by $a_{\text{THz}} = \Gamma_{\text{THz}} \times a_{\text{DPM}} = 5.0 \times 10^{10} \times 3.772 \times 10^{-57} = 1.886 \times 10^{-46} \text{ m/s}^2$

6. Astrophysical Significance

Discovery: This is the first UQFF derivation showing that the DPM vacuum coupling leaves a measurable time-signature as an SNR expands. The gravity signal decreases monotonically as $r(t)^{-3}$, encoding the full expansion history since SN 1054. Observational correlates include:

1. **Hubble Space Telescope (HST):** Crab optical wisp proper motion ~ 0.015 arc-sec/yr confirms $v_{\text{exp}} \approx 1500$ km/s at the shock front (consistent with $v_{\text{exp}} = 1.5 \times 10^6$ m/s)
 2. **Chandra X-ray Observatory:** Ring/jet variability on 1-3 year timescales provides empirical window into the DPM vacuum coupling variation as $V_{\text{sys}}(t)$ changes
 3. **PAPER_290 Prediction:** At the same observing epoch, a radio-quiet SNR of identical age but lower v_{exp} would show a higher a_{DPM} — the dilution is purely geometric ($V(t)$ scaling)
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7. Relationship to Prior UQFF Papers

- **PAPER_287 (Session 81 RSC):** Established fixed-volume DPM ($V_{\text{sys}} = \text{constant} = 4.189 \times 10^{12} \text{ m}^3$)
 - **PAPER_290 (This paper):** FIRST dynamic $V_{\text{sys}}(t)$ — the DPM signal evolves as $V(t)^{-1}$
 - The V_{sys} in RSC ($4.189 \times 10^{12} \text{ m}^3$, neutron star $\sim r=10^4$ m) is 12 orders of magnitude smaller than the Crab V_0 ($5.889 \times 10^{50} \text{ m}^3$), explaining why RSC a_{DPM} ($3.545 \times 10^{-18} \text{ m/s}^2$) is 38 orders of magnitude larger than Crab a_{DPM} ($3.772 \times 10^{-57} \text{ m/s}^2$)
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8. Wolfram KB Registration

CRAB_UQFF: $a_{\text{DPM}}(t) = F_{\text{DPM}} \cdot f_{\text{DPM}} \cdot E_{\text{vac}} / (c \cdot (4/3) \cdot \pi \cdot (r_0 + v_{\text{exp}} \cdot t)^3)$
 $F_{\text{DPM}} = 6.284 \times 10^{26} \text{ N}$; $a_{\text{DPM}}(0) = 2.521 \times 10^{-56} \text{ m/s}^2$; $a_{\text{DPM}}(971 \text{ yr}) = 3.772 \times 10^{-57} \text{ m/s}^2$; $D = 6.69 \times$

$\text{Gamma_THz} = 10 * f_THz * v_exp / c = 5.0e10$ (1500x RSC) [PAPER_290 SNR DPM Dilution]

Session 82 — 24th C++ UQFF Module — PAPER_290 of 1000